

Hamdard University Bangladesh

Faculty of Science, Engineering & Technology
Mid-Term Examination (Spring '23)
PHY 121 Physics II

Time: 1.5 hours

Marks: 30

Answer any 3 (Three) from the following sets of questions

1. (a) Write down the short notes on Point Charge, Conductor and Insulator. 3
(b) States and derive the Coulomb's law. 3
(c) Two charges (A & B) are 2.5 mm apart. Charge A has a charge of $4 \mu\text{C}$ while charge B has a charge of $-2 \mu\text{C}$. What is the electrostatic force between two charges? 4
2. (a) Define capacitance. Calculate the equivalent capacitance of three series capacitor, where $C_1 = 4\text{C}$, $C_2 = \text{C}$ and $C_3 = 2\text{C}$ 5
(b) Derive the equation of energy stored in a capacitor with suitable figure. 5
3. (a) Write down the statement of Gauss's law. 3
(b) Deduce the Coulomb's law by using Gauss's law 7

4. (a) Find the electric field (E) in point P. Distance from P point to all charges is 30cm and $Q_1 = +2 \mu\text{C}$, $Q_2 = +12 \mu\text{C}$ and $Q_3 = -8 \mu\text{C}$. 8

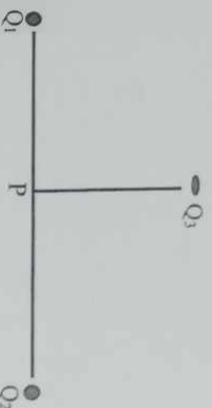


Fig. 4(a)

- (b) What is Electric Field?

2



Answer any three (03) out of the following questions:

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

3 × 10 = 30

1. (a) Amplify the statement "look before you leap" in not more than 200 words. [6]
(b) Describe the things that you have to keep in mind while delivering a presentation. [4]

2. Write an essay on 'Laptop' in not more than 500 words. Your essay must cover a laptop's (i) definition, (ii) types, (iii) major manufacturing companies, (iv) uses, (v) demerits [10]

3. (a) Expound on the appropriate uses of CC and BCC in writing and sending an email. [4]
(b) Amplify the statement "don't bite the hands that feed you" in not more than 200 words. [6]

4. (a) What is a 'terminator' in a paragraph? Generate a topic sentence for a paragraph titled 'Bollywood'. [2]
(b) The following paragraph is incoherent. Provide a suitable title for the paragraph and rewrite it coherently: [8]

Also the city of Comilla was once the capital of Epara kingdom. Along with the Comilla Economic Zone EPZ, various industrial factories have come up at the airport, making the airport area a business and commercial city. Comilla officially spelled Cumilla is a metropolis on the banks of the Gomti River in the southeastern tip of Bangladesh. It was one of the cities of ancient Bengal. The area of Comilla City Corporation is 53.04 square kilometers, so the surrounding areas of the main city fall under the jurisdiction of the City Corporation. Comilla Airport is located in the Dulipara area of Comilla city. It is a district, of which, its citizens can be duty proud. Comilla City is a division centered city district with surrounding districts coming to Comilla in more diverse areas of work. Besides, Bibi Bazar land port has come up 5 km away from Comilla city. The urban areas falling outside the city corporation are considered as suburbs with a population of 6 lakhs.

5. (a) Suppose that your group is delivering a 30-minutes oral presentation on the topic 'the Internet'. Write a 6-minutes script on the sub-topic 'Children and the Internet'. [6]
(b) Suppose that you are in charge of collecting the fees from your batch for the upcoming freshers' reception ceremony of your department. Write an email to the Head of your department notifying him/her about the progress of your task. [4]

Answer any three questions.

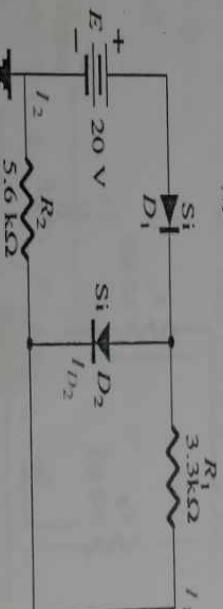
Marks: 10 x 3 = 30

1. (a) Define PN junction diode. Define knee voltage and breakdown voltage of diode with example(s). 03
- (b) What is Zener diode? Why is it used? 03
- (c) Explain PN junction acts as a full wave rectifier and a half wave rectifier with example(s). 04

2. (a) For the following series diode configuration, Determine V_D , V_K , and I_D . 04

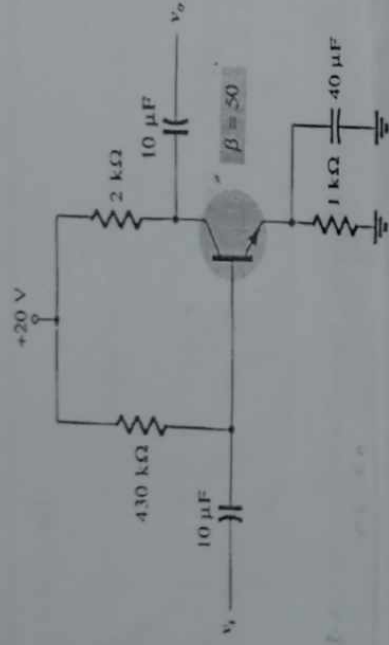


- (b) Determine the currents I_1 , I_2 and I_{D2} from the given figure 06



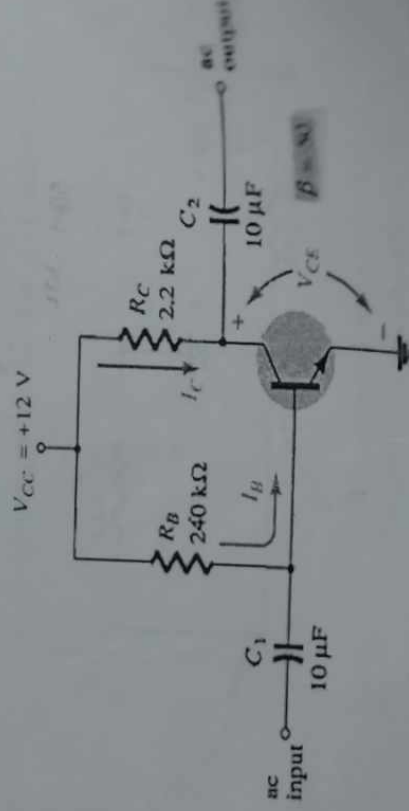
3. (a) Define BJT. Explain the types and structure of BJT. 04
- (b) Explain transistor working principle with figure(s). 06
4. (a) Define NPN and PNP transistor. State are the differences between them. 03

- (b) For the following emitter-base connection, determine I_B , I_C , V_{CE} , V_C , V_B , V_E , V_{BC} and V_{BE} : 07



- (a) What happens to the depletion region while the diode is in forward bias and reverse biases? 03

- (b) Determine I_{BQ} , I_{CQ} , V_{BEQ} , V_B , V_C , and V_{BC} for the below configuration : 07





HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
MAT129: Linear Algebra and Complex Variables
Mid Term Examination
7/8/2023

Total Time: 1 hour 30 mins.

Semester: Spring 2023

Total Marks: 30

Answer any Three (03) of the following questions:
[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

1. (a)

Solve the system of linear equations

$$3 \times 10 = 30$$

$$\begin{aligned}x_1 + 2x_2 - x_3 + 3x_4 &= 3 \\2x_1 + 4x_2 + 4x_3 + 3x_4 &= 9 \\3x_1 + 6x_2 - x_3 + 8x_4 &= 10\end{aligned}$$

[5]

(b)

Find the Rank of the Matrix $A =$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 5 & 6 & 7 & 4 \\ 4 & 7 & 10 & 13 & 16 \\ 5 & 8 & 9 & 10 & 3 \end{pmatrix}$$

[5]

2. (a)

Define linear dependence & independence. Explain where the following vectors in $\mathbb{R}^3$ are linearly dependent or independent. Where vectors are $(1, 1, -1), (1, 2, 3), (4, 5, -3)$.

[6]

(b)

Prove that, suppose two or more non zero vectors $v_1, v_2, \dots, v_m$ are linearly dependent. Then one of them is a linear combination of preceding vectors.

[4]

3. (a)

Find the inverse of the matrix $A = \begin{pmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{pmatrix}$

[5]

(b)

Define linear combination. Determine whether or not the vector $v = (2, -5, 3)$ is a linear combination of the vectors $u_1 = (1, -3, 2), u_2 = (2, -4, -1), u_3 = (1, -5, 7)$.

[5]

4. (a)

Solve the systems of equations by Gauss - Jordan Elimination method

[5]

$$\begin{aligned}3x - 2y + 8z &= 9 \\-2x + 2y + z &= 3 \\x + 2y - 3z &= 8\end{aligned}$$

(b)

Prove that $\begin{vmatrix} 1 & a & a^2 & 0 \\ 0 & 1 & a & a^2 \\ a^2 & 0 & 1 & a \\ a & a^2 & 0 & 1 \end{vmatrix} = 1 + a^4 + a^8$

[5]

5. (a)

Define basis and dimension. Determine whether or not the following sets form a basis for the vector space in $\mathbb{R}^3$, $\{(1, 2, 0), (0, 5, 7), (-1, 1, 3)\}$.

[5]

(b)

What is linear transformation? Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$. Then find $\text{Ker } T$ and $\text{Im } T$.

[5]

```

int main() {
    int num1 = 5;
    int num2 = 3;
    multiplyNumbers(num1, num2);
    return 0;
}

```

- (a) Outline the practical uses of C programming that have a direct impact on people's lives.
- (b) How does C achieve portability across different platforms?
- (b) Sketch a diagram illustrating the key components and organization of a C program.

1 ✓
2 ✓
3 ✓

- (a) How one can effectively implement compartmentalization within a C program?
- (b) Which escape sequences should be familiar to C programmers?
- (c) What is the output of the given code?
- ```

int a = 4;
int b = 2;
int result = a << b;
printf("%d", result);

```

2 ✓  
2 ✓  
2 ✓

- (a) What output will be displayed by the following code?

```

int a = 5;
int b = 2;

```

```

float result = (float)a / b;

```

```

printf("%f", result);

```

- (b) What will be the output of the following C program?

```

#include <stdio.h>
int main() {

```

```

 int i = 3;
 int *j;

```

```

 j = &i;
 j++;

```

```

 printf("%d", *j);
 return 0;
}

```

8421  
0100  
20010  
32 16 8 4 2  
- 0 1 0 0 0  
switche  
swita



# HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology  
Department of Computer Science and Engineering  
CSE 121: Structured Programming Language

Batch: 23<sup>rd</sup>

Date: 19/06/2023

## Mid Term Examination

Semester: Spring 2023

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the following:

5 x 6 = 30

✓ (a) In what ways do high-level, assembly, and machine language differ from each other? 3

(b) Write the source code for the following algorithm: 3

1. Start
2. Read n
3.  $i = 0$ , sum = 0
4.  $i = i + 1$
5. sum = sum + ( $i * i$ )
6. Repeat steps 4 and 5 until  $i > n$
7. Print sum
8. Stop

2 (a) Compare and contrast fixed and variable length arrays based on their array lengths. 3

(b) Write a C program to find the sum of all elements of a given array. 3

✓ 3 (a) Convert the following code of while loop with other loops. 3

#include <stdio.h>

int main()

{

int i;

while(0, i < 4)

{

printf("Loop ");

i++;

}

return 0;

}

requires

do {  
i++  
while (i < 4)

(c) What will be the output of the following C program?

```
#include <stdio.h>
int main() {
 char str[] = "Hello, World!";
 int i = 0;
 while (str[i]) {
 if (str[i] == 'o') {
 str[i] = 'O';
 }
 i++;
 }
 printf("%s", str);
 return 0;
}
```



# Hamdard University Bangladesh

Faculty of Science, Engineering & Technology

Final Examination (Spring'23)

PHY 121 Physics II

Marks: 40

Time: 2 Hours

Answer any 4 (Four) from the following sets of questions

- 1 a) Define energy density. Determine energy density of electric field in a capacitor. 4
- b) A plane parallel plate capacitor has circular plates of radius  $r = 10$  cm, separated by a distance  $d = 1$  mm. (i) How much charge is stored on each plate when their electric potential difference has the value 100 V? (ii) Calculate the electric field, the electric field density and energy stored in the capacitor. 6
- 2 a) State the Biot-Savart law. 3
- b) Define magnetic field. Determine magnetic field due to a circular current carrying coil. 7
- 3 a) State the Ampere's law. 2
- b) Derive Ampere's law using Biot-Savart law 3
- c) Two long and fixed parallel wires, A and B are 10 cm apart in air and carry currents of 40 and 20 A, respectively in same directions. Determine resultant flux density (i) on a line midway between the wires and parallel to them and (ii) on a line 18 cm from wire B and 8 cm from wire A. 5
- 4 a) Write down the short notes on Current density, Ohm's law, KVL, KCL and electron drift velocity. 5
- b) You are given a length of nichrome wire having length  $l = 10$  m and resistance  $R = 23 \times \text{your id last 2 digit} \Omega$ . It has to be connected across a 120 V line. Under which circumstances will the wire dissipate more heat (i) its entire length is connected across the line or (ii) the wire is cut in half and the two halves are connected in parallel across the line? 5
- 5 a) Explain the statement of Faraday's electromagnetic induction law and Lenz's law 3
- b) Determine magnetic field near a long current carrying wire using Ampere's law. 3
- c) A solenoid 0.5 m long has 2000 turns. The magnetic induction near the centre of the solenoid is 0.08 T. What is the current in the solenoid?  $B = \frac{\mu_0 N \cdot I}{2l}$  4
- 6 a) Write down the two postulates upon which the special theory of relativity is based. 3
- b) Derive the formula for relativistic variation of mass with velocity. 7

- (a) What is recursion in programming? 3
- (b) Imagine you have a set of Matryoshka dolls, also known as Russian nesting dolls. These dolls come in a series of decreasing sizes, with each doll fitting inside a larger one. The smallest doll is placed inside the next larger doll, and so on, until you reach the largest doll. Each doll can be seen as a container that holds a smaller version of itself. Now, find how the concept of recursion in programming is reflected in this scenario. 3

(c) Write a recursive function to calculate the factorial of a positive integer. 3

- (a) How do you release memory back to the system using the free function? Provide an example. 2

- (b) Find the error in the code and provide a detailed explanation of why it's incorrect. How can you fix the error to achieve the intended result? 3

```
#include <stdio.h>
int main() {
 int a = 5, b = 10;
 int c = a b;
 printf("c: %d\n", c);
 return 0;
}
```

- (c) Identify the error in the code, explain the scope-related problem, and propose a solution to achieve the desired output. 3

```
#include <stdio.h>
void modifyValue(const int num) {
 num = num * 2;
}
int main() {
 int x = 5;
 modifyValue(x);
 printf("Modified value: %d\n", x);
 return 0;
}
```

- (a) What is a macro in C, and how does it differ from a function? 2
- (b) Explain how you can use the `__FILE__`, `__LINE__`, `__DATE__`, and `__TIME__` macros together to create helpful debugging messages. 2



# HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology  
Department of Computer Science and Engineering  
CSE 121: Structured Programming Language

Batch: 23<sup>rd</sup>

Date: 05/09/2023

Semester: Spring 2023  
Time: 2 hour  
Total Marks: 40

## Final Examination

Answer any five questions from the following:

5 X 6 = 30

- 1 (a) How do you define a structure? Provide an example of a structure declaration. 3
- (b) Write a C program that defines a "Value" union containing "integer" and "float" members. Assign values to both members and demonstrate how accessing one affects the other. 5
- 2 (a) Explain how you can assign values to enumeration constants. 3
- (b) Write a C program that defines an enumeration for different genres of books (Mystery, Sci-Fi, Fantasy) and uses typedef to create a type alias for the enumeration. Display the list of book genres using the alias. 5
- 3 (a) How do you open a file for reading or writing in C? Provide an example of opening a file in both modes. 4
- (b) Explain the difference between text files and binary files. When might you choose one over the other? 4
- 4 (a) Write a program that rounds down a user-entered floating-point number to the nearest integer using the floor function. 2
- (b) How do you use typedef to create an alias for a data type? Provide an example. 2
- (c) Write a C program that takes the radius of a circle as input from the user. The program should then calculate and display both the area and the perimeter (circumference) of the circle using appropriate math functions. Use the value of  $\pi$  as 3.14159 for calculations. 4

Example output:

Enter the radius of the circle: 5.2

Circle Area: 84.948912





Read the following text and complete the tasks that follow:

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

The central processing unit (CPU) is the primary component of a computer system that carries out the instructions of a computer program by performing basic arithmetic, logical, control, and input/output (I/O) operations. It is often referred to as the "brain" of the computer.

The CPU consists of several key components that work together to process information and execute instructions. The main function of the CPU is to receive, interpret, and execute instructions from the computer's memory, enabling the computer to perform various tasks.

At its core, a CPU is an electronic circuit that operates on binary data, which consists of ones and zeros. These binary digits, also known as bits, represent the most basic form of information in a computer system. The CPU processes these bits in a sequential manner, following a predefined set of instructions called machine code.

### Components of a CPU

A CPU consists of several key components that work together to process and execute instructions. These components include the control unit, arithmetic logic unit (ALU), cache memory, registers, clock speed, and the pipeline. Let's take a closer look at each of these components:

1. **Control Unit:** The control unit is responsible for controlling the flow of data and instructions inside the CPU. It fetches instructions from the computer's memory, decodes them, and manages the execution of those instructions by coordinating with other components of the CPU.
2. **Arithmetic Logic Unit (ALU):** The ALU performs all the mathematical and logical operations in a CPU. It can perform tasks such as addition, subtraction, multiplication, division, and logical comparisons. The ALU processes data based on the instructions received from the control unit.
3. **Cache Memory:** Cache memory is a small, high-speed memory located inside the CPU. It stores frequently accessed data and instructions to provide quick access for the CPU. By storing this information closer to the processor, cache memory helps reduce the time it takes to fetch data from the computer's main memory.
4. **Registers:** Registers are small, high-speed storage locations within the CPU. They are used to store temporary data and instructions that the CPU needs immediate access to. Registers play a crucial role in the execution of instructions, as they enable faster data manipulation and processing.
5. **Clock Speed:** The clock speed of a CPU determines how many instructions the CPU can execute per second. It is measured in hertz (Hz) or gigahertz (GHz). A higher clock speed indicates a faster CPU, as it can complete more instructions in a given amount of time. However, the efficiency of a CPU is not solely determined by its clock speed but also by other architectural factors.
6. **Pipelining:** Pipelining is a technique used in CPUs to increase their efficiency. It involves dividing the execution of instructions into multiple stages and allowing the CPU to process different stages of multiple instructions simultaneously. This overlap of operations improves throughput and overall performance.



(c) Identify the type of error present in each of the following code snippets. 4 X 1 = 4

i. 

```
#include <stdio.h>
int main() {
 printf("Hello, World!\n");
 return 0;
}
```

*syntax*

ii. 

```
#include <stdio.h>
int main() {
 int radius = 5;
 double area = 3.14 * radius * radius; // Incorrect formula for
 area of a circle
 printf("Area: %f\n", area);
 return 0;
}
```

*Logical*

iii. 

```
#include <stdio.h>
int main() {
 int x;
 printf("Value of x: %d\n", x);
 return 0;
}
```

*uninitialization*

iv. 

```
#include <stdio.h>
int main() {
 int numerator = 10;
 int denominator = 0;
 int result = numerator / denominator;
 printf("Result: %d\n", result);
 return 0;
}
```

~~memory allocation~~  
*runtime*



1. Briefly answer the following questions.

- Why is the CPU equated with a brain?
- What does 'bits' refer to?
- How does Cache Memory lessen the data-fetching time?
- Where are the short-term data and instructions reserved?
- Which is the indicator of a faster or slower CPU?

1x5=5

2. Judge the following statements as true or false. If false, mention the true one.

1x5=5

- A CPU is indivisible into separate components.
- The control unit maintains the flow of external data.
- The mathematical functions are carried out by the ALU.
- Hertz is a unit of time.
- The technique to better the orderliness of the CPU is pelining.

3. Match the words from column A with their meanings from column B.

1x5=5

| A               | B                                                                     |
|-----------------|-----------------------------------------------------------------------|
| i) logic        | i) forming or following in a chronological order                      |
| ii) interpret   | ii) cause (something) to occur in a particular way                    |
| iii) sequential | iii) conducted or assessed according to strict principles of validity |
| iv) determine   | iv) put (a plan, order, or course of action) into effect              |
| v) execute      | v) extend over so as to cover partly                                  |
|                 | vi) explain the meaning of something                                  |

4. Summarize the passage in not more than eight sentences.

5

5. Detect the parts of speech of any ten of the following words.

0.5x10=5

- primary; operation; execute; known; flow; performance; each; closer; based; manipulation; frequently; enable; however; dividing; quick.

6. Rewrite the following passage with appropriate articles where necessary.

0.5x10=5

Electrical engineering is an engineering discipline concerned with study, design, and application of equipment, devices, and systems which use electricity, electronics, and electromagnetism. It emerged as an identifiable occupation in the latter half of the 19th century after the commercialization of the electric telegraph, telephone, and electrical power generation, distribution, and use. Electrical engineering is now divided into a wide range of different fields, including computer engineering, systems engineering, power engineering, teletronics, and optics and photonics. Many of these disciplines overlap with other engineering branches.

7. From the following pronunciations, detect any four words from the passage.

0.5x4=2

- a) 'sentrel ; b) 'vfn ; c) kam'pjuto ; d) 'mem'ri ; e) 'dento ; f) klbk

8. Rearrange the following sentences in appropriate order.

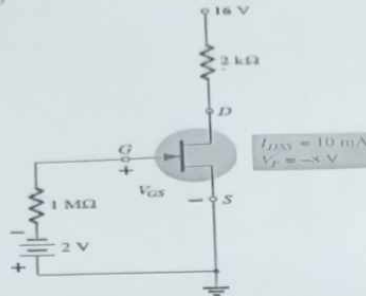
1x8=8

- The belligerents involved are 120 Indian soldiers accompanied by four Hunter fighter aircraft and 2,000-3,000 Pakistani soldiers accompanied by 30-40 tanks.
- On the other hand, Pakistan lost 200 soldiers, 36 tanks, and more than 500 vehicles.
- The Battle of Longewala (4-7 December 1971) was one of the first major engagements in the western sector during the Indo-Pakistani War of 1971.
- The 1997 Bollywood film *Border*, which was directed by J. P. Dutta and starred Sunny Deol as Major Kuldeep Singh, depicts this battle.
- It was fought between assaulting Pakistani forces and Indian defenders at the Indian border post of Longewala, in the Thar Desert of Rajasthan.
- The Indian casualties and losses include 2 soldiers, 1 M40 recoilless rifle, and 5 camels.
- The war resulted in Indian victory.
- The Pakistani commanders made several questionable decisions in the battle.



5. (a) Draw the symbols of the n-channel and p-channel JFETs. 02  
(b) Determine the following for the self-bias configuration network: 06

- i.  $V_{GS0}$
- ii.  $I_{DQ}$
- iii.  $V_{DS}$
- iv.  $V_S$
- v.  $V_0$
- vi.  $V_D$

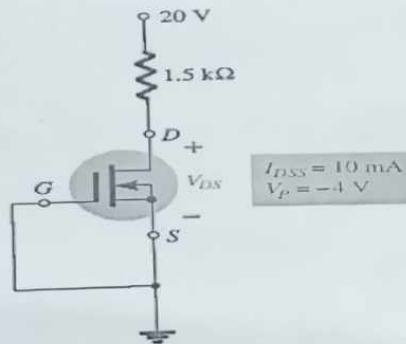


6. (a) Write the symbol of MOSFET for n-channel and p-channel. 02  
(b) What is the meaning of Depletion mode and Enhancement mode in MOSFET? 02  
(c) Find the drain current with the following parameters: 04

$$I_{DSS} = 5 \text{ mA}, V_{GS} = -2 \text{ V}, V_P = -1 \text{ V}$$

7. (a) Draw MOSFET output characteristics curve between  $V_{DS}$  and  $I_D$  with proper designation. 03

- (b) Determine  $I_{DQ}$  and  $V_{DS}$  for the below diagram: 05





# HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology  
Department of Computer Science and Engineering  
EEE127: Electronic Devices & Circuits

Batch: 23<sup>rd</sup>

Date: 09/09/2023

Final Examination

Semester: Spring 2023

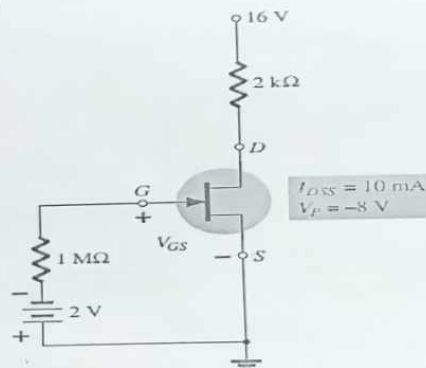
Time: 02:00 Hours

Total Marks: 40

Answer any five questions.

Marks: 8 x 5 = 40

- ✓ (a) What is a Field Effect Transistor (FET)? 01
- (b) Write the similarities and differences between FETs and BJTs. 03
- (c) Explain JFET operating characteristics for pinch-off voltage with necessary diagram. 04
- ✓ 2. (a) Explain the working principle of the JFET when  $V_{GS} = 0V$  and  $V_{DS}$  is increased from 0 (zero) to a more positive voltage. 04
- (b) Draw and explain the JFET operating characteristics saturation curve between  $I_D$  and  $V_{DS}$ . 04
3. (a) What are operational amplifiers (Op-amps)? Why use op-amps? 02
- (b) Define an oscillator. Why do we need to study semiconductors? 03
- (c) Write the various applications of semiconductors in detail with explanation. 03
- ✓ 4. (a) Write the three basic operating conditions for a JFET. 02
- (b) Determine the following for the fixed-bias configuration network: 06
  - i.  $V_{GSQ}$
  - ii.  $I_{DQ}$
  - iii.  $V_{DS}$
  - iv.  $V_D$
  - v.  $V_G$
  - vi.  $V_S$







Answer any four (04) of the following questions:

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

4 × 10 = 40

1. (a) What is complex number and complex plane? [02]  
(b) Describe the region determined by  $|z - 2| - |z + 2| > 3$ . [04]  
(c) For any complex numbers  $z_1, z_2, z_3, \dots, \dots, z_n$ . Then show that  $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$  [04]
2. (b) Define an analytic function. Show that Cauchy's Riemann equations are satisfies for  $e^{z^2}$ . [05]  
(c) State and prove the necessary conditions for a function  $f(z)$  to be analytic in a region. [05]
3. (a) Define harmonic conjugate function and show that the function  $u = x^3 + 6x^2y - 3xy^2 - 2y^3$  is a harmonic function. Also find harmonic conjugate of  $u$ . [05]  
(b) Solve the following equations: [02]  
(i)  $x^6 - 3x^3 + 3 = 0$  [03]  
(ii)  $(x + 1)^5 + (x - 1)^5 = 0$  [01]  
4. (a) What is complex integration? [05]  
(b) State and prove Cauchy's integral formula [04]
5. (c) Find  $\oint_C \frac{\sin 3z}{(z + \frac{\pi}{2})} dz$ , where  $C$  is the circle  $|z| = 5$ . [04]  
(a) Define singular point, isolated singular point and pole. [02]
- (b) Locate and name the singularities in the finite  $z$  plane of the function  $f(z) = \frac{z}{(z^2 + 9)^2}$  [03]
6. (c) If we choose the principal branch of  $\sin^{-1}z$  to be that one for which  $\sin^{-1}0 = 0$ . Prove that  $\sin^{-1}z = \frac{1}{i} \ln(iz + \sqrt{1 - z^2})$  [05]  
(a) State Taylor and Laurent series. [02]
- (b) Determine the Laurent series for  $f(z) = \frac{1}{z+6}$  that are valid in the region  
(i)  $|z| < 6$  [04]  
(ii)  $|z| > 6$
- (c) Define contour and solve the contour integration  $\int_2^{4-2i} (z^2 - 4iz + 6) dz$  [04]